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LOW-VELOCITY IMPACT DAMAGE ASSESSMENT AND NUMERICAL MODELLING OF GLASS/PPS LAMINATED COMPOSITES USING COMMERCIAL FINITE ELEMENT TOOLS

F aculty of Engineering of University of Porto

MASTER IN MECHANICAL ENGINEERING – AERONAUTICAL AND VEHICLES STRUCTURES

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IMPACT DAMAGE MECHANISMS

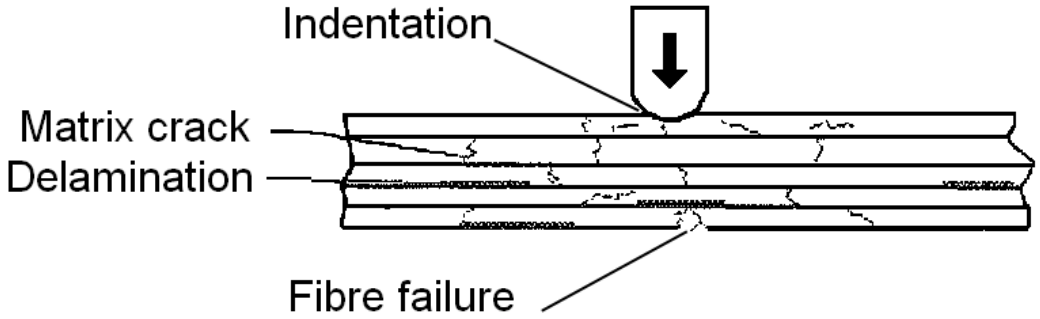


FIGURE 1.1 | Typical impact damage mechanisms in composite laminates [1].

- Composite laminates widely used in **aerospace structures**
- Low-velocity impacts generate **internal damage**
- Internal damage may significantly reduce **structural integrity**

OBJECTIVES OF THIS WORK

- Develop progressive **numerical models** for impact in Glass/PPS laminates
- Investigate the influence of **different damage modelling strategies**
  (Elastic → Hashin → Cohesive → Bi-phasic)
- Identify the **damage mechanisms governing impact response**
- Compare numerical predictions with **experimental and NDT results**

PRESENTATION STRUCTURE

Experimental Basis

- Impact test Campaign

Numerical Model

- Finite Element Model

Modelling Strategies

- Elastic Impact Model
- Intralaminar Damage (Hashin)
- Interlaminar Damage (Cohesive)
- Bi-phasic Modelling

Validation

- Post-Impact NDT

Closing

- Conclusions

EXPERIMENTAL IMPACT CAMPAINGN

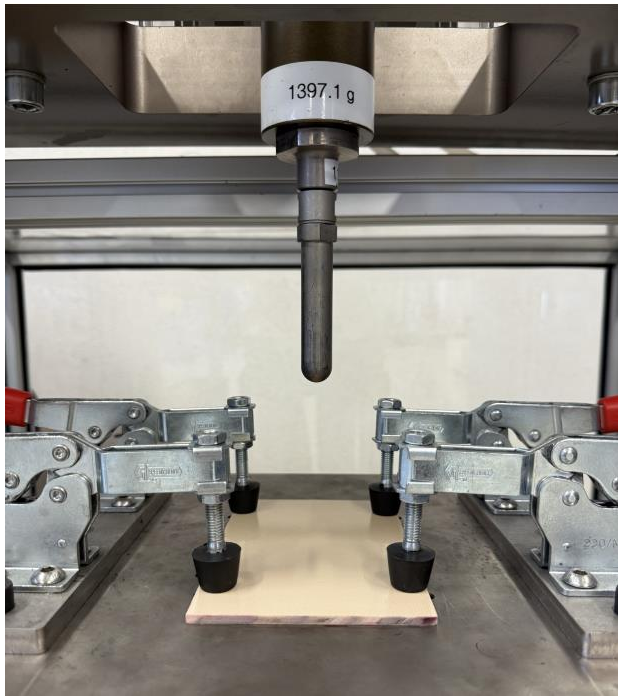


FIGURE 2.1 | Drop-weight impact setup used in the experimental campaign [2].

TABLE 2.1 | Summary of experimental impact cases and measured damage metrics.

Impact Energy [J]	Indentation [mm]	Superficial Crack Length [mm]	No. Specimens
4.5	3.9	0.0	3
7.5	4.8	3.2	3
14.2	7.4	10.3	3
27	9.0	19.0	1

- Experimental campaign performed in previous work *(Martina D'Aiello) [2]*
- Drop-weight impact tests on **Glass/PPS woven laminate** (5HS satin weave)
- **Layup:** $[0/90]_8s$
- **Plate size:** 148 × 100 × 3.7 mm
- Four impact energy levels investigated *(4.5–27 J)*

FINITE ELEMENTO MODEL OVERVIEW

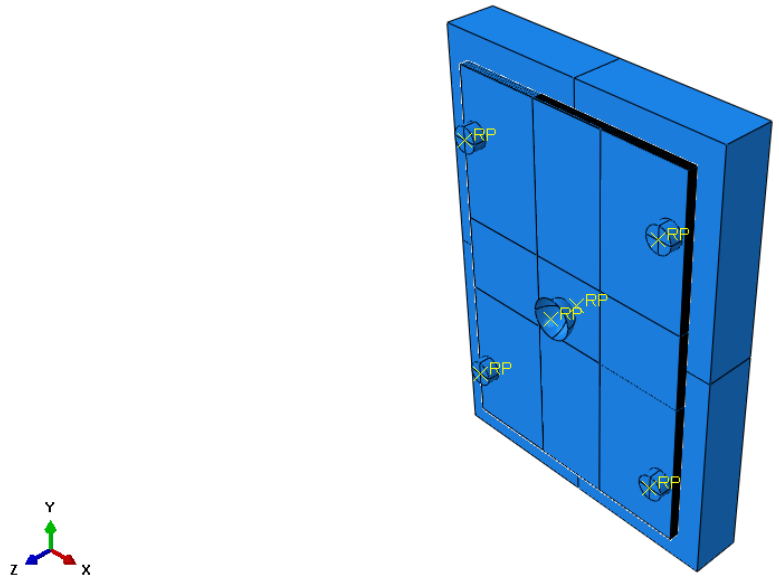


FIGURE 3.1 | *Finite Element Model Assembly Model.*

- Finite element model developed in **Abaqus/Explicit**
- Model reproduces the **experimental drop-weight configuration**
- Impactor modelled as **rigid body**
- Composite laminate discretized using **continuum shell elements (SC8R)**

BOUNDARY CONDITIONS

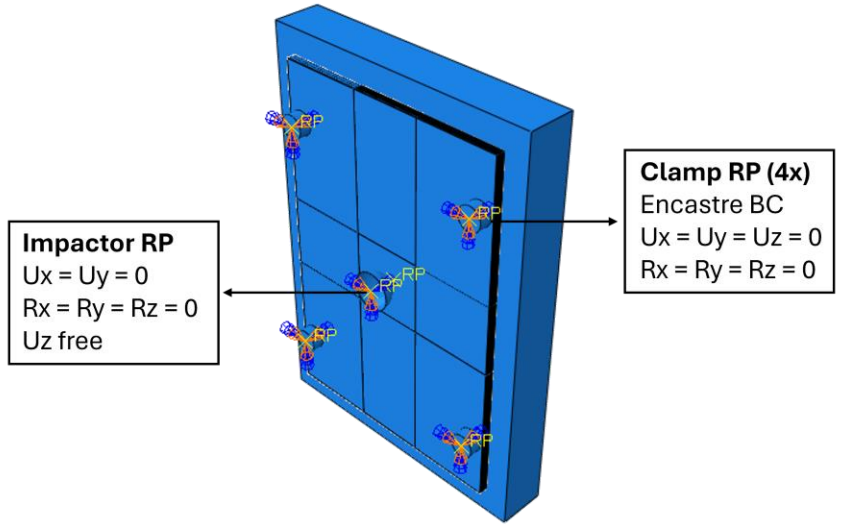


FIGURE 3.2 | *Clamp constraints and impactor control.*

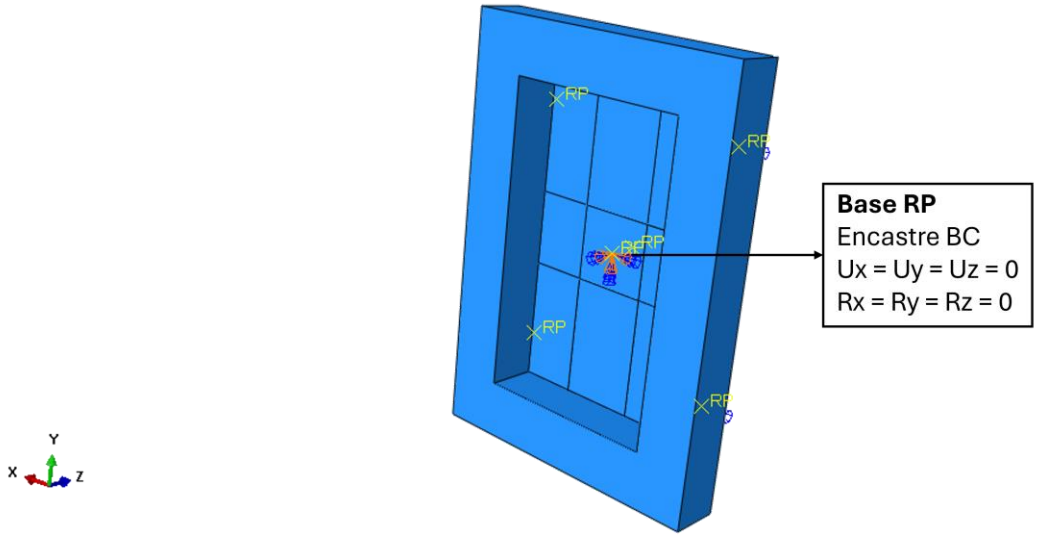


FIGURE 3.3 | *Base support boundary condition.*

- Boundary conditions reproduce the **experimental fixture**
- Specimen supported using **base and clamp constraints**
- Impactor controlled through a **reference point (RP)**
- Impact generated by the applied **initial velocity**

MESH AND MODEL DISCRETIZATION

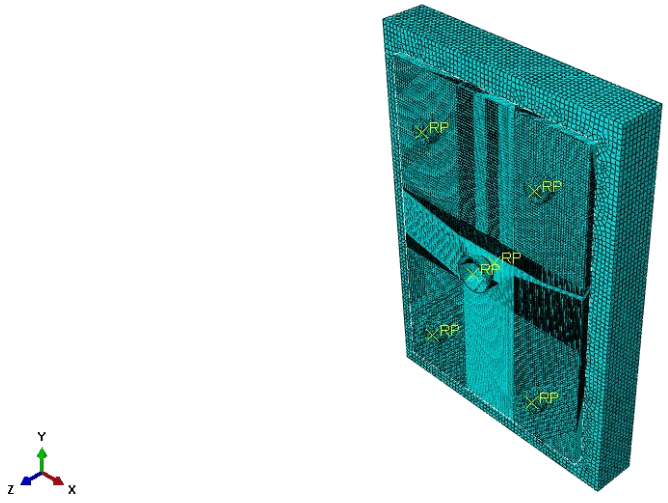


FIGURE 3.4 | Global finite element mesh of the impact model.

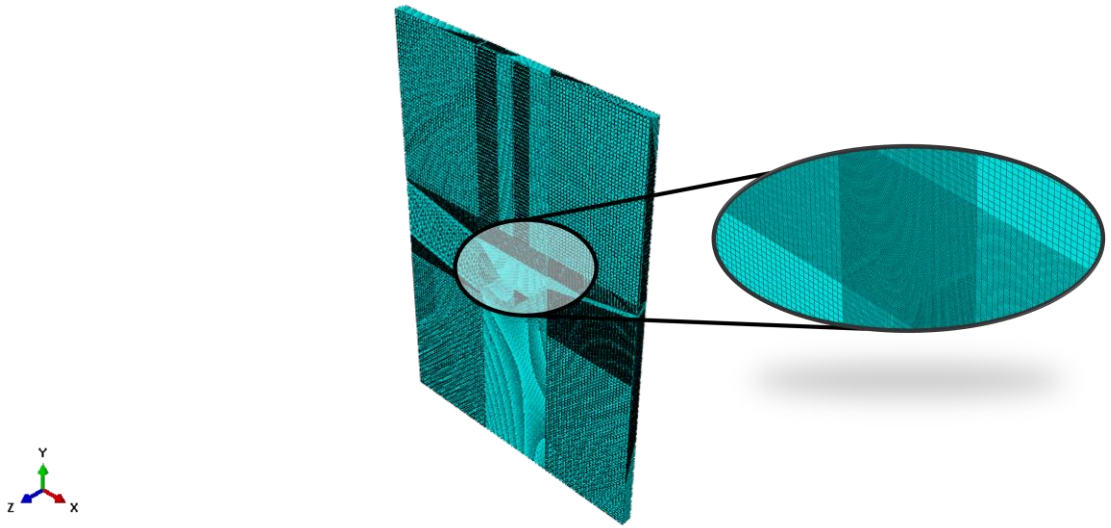


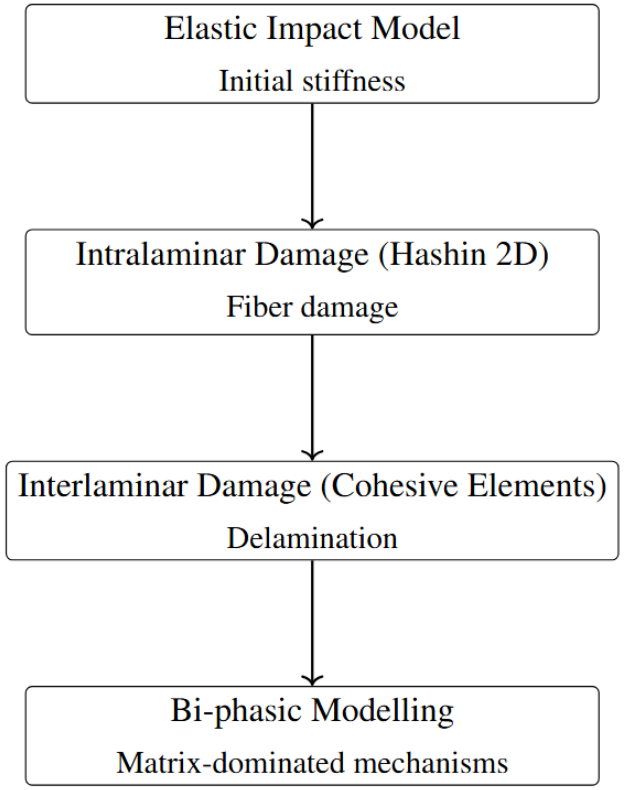
FIGURE 3.5 | Local mesh refinement in the impact region.

TABLE 3.1 | Finite element discretization of the numerical model.

Component	Element	Elements
Composite panel	SC8R	705280
Rubber clamps	C3D8R	3360
Supporting base	C3D8R	32270
Impactor	R3D4	9184
Total	–	**750094**

- Structured mesh with **local refinement in the impact zone**
- Composite laminate discretized using **continuum shell elements (SC8R)**
- Total model size: **~750k elements**

PROGRESSIVE MODELLING STRATEGY



- **Progressive** inclusion of the governing impact damage mechanisms

FIGURE 4.1 *| Progressive development of the numerical modelling strategy.*

ELASTIC MODEL: MATERIAL PROPERTIES

TABLE 4.1 | Orthotropic elastic properties of the composite laminate [3].

Property	Value
Young's modulus (warp)	$E_1 = 22.49$ GPa
Young's modulus (weft)	$E_2 = 19.29$ GPa
Through-thickness modulus	$E_3 = 6.10$ GPa
Poisson's ratio	$\nu_{12} = 0.113$
Poisson's ratio	$\nu_{13} = 0.32$
Poisson's ratio	$\nu_{23} = 0.32$
In-plane shear modulus	$G_{12} = 3.60$ GPa
Transverse shear modulus	$G_{13} = 3.00$ GPa
Transverse shear modulus	$G_{23} = 3.00$ GPa
Density	$\rho = 1.95$ g/cm^3

TABLE 4.2 | Elastic properties of the rubber clamps and steel base [4,5].

Property	Rubber clamps (SBR)	Steel base
Young's modulus, E [GPa]	6	210
Poisson's ratio, ν [-]	0.10	0.30
Density, ρ [g/cm^3]	0.94	7.80

ELASTIC MODEL: EXPERIMENTAL COMPARISON

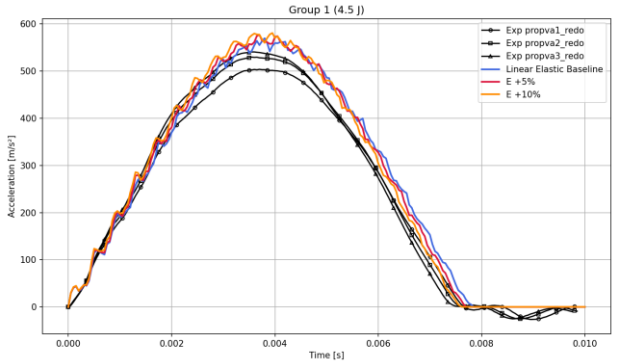


FIGURE 4.2 | *Experimental vs numerical response (4.5 J) - Elastic.*

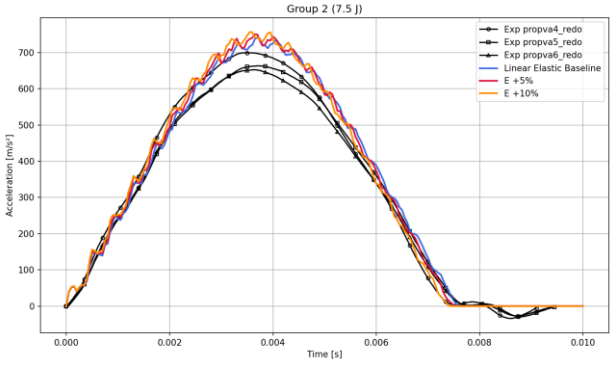


FIGURE 4.3 | *Experimental vs numerical response (7.5 J) - Elastic.*

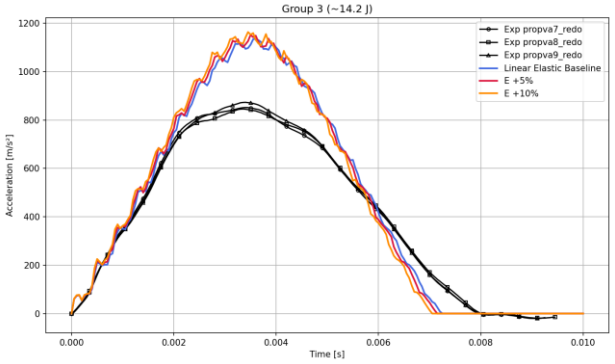


FIGURE 4.4 | *Experimental vs numerical response (14.2 J) - Elastic.*

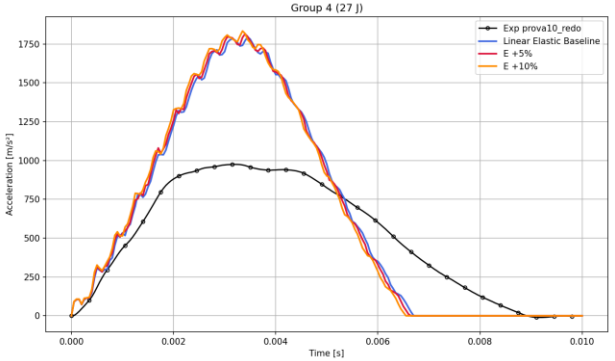


FIGURE 4.5 | *Experimental vs numerical response (27 J) - Elastic.*

ELASTIC MODEL: STIFFNESS SENSITIVITY

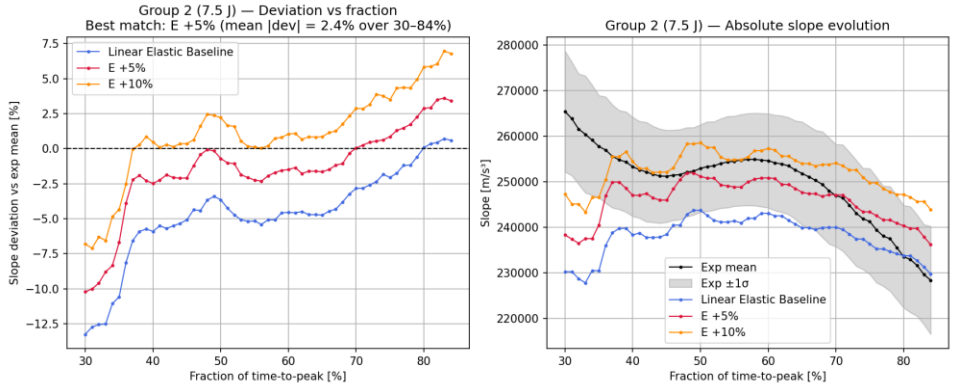


FIGURE 4.6 | Model stiffness sensitivity to E_1 and E_2 (7.5 J) - Elastic.

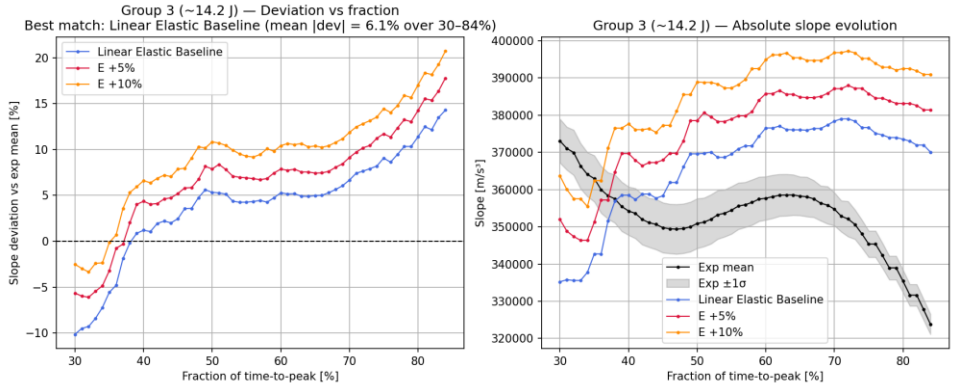


FIGURE 4.7 | Model stiffness sensitivity to E_1 and E_2 (14.2 J) - Elastic.

- **Baseline** model slightly underestimates stiffness
- Increasing **E_1,E_2** improves agreement with experiments
- **Best compromise obtained with E +5%**

INTRALAMINAR DAMAGE MODELLING

- Intralaminar damage modelled using the **Hashin 2D failure criteria**
- **Plane-stress formulation** (Abaqus built-in composite damage model)
- Suitable for **thin laminate configurations**
- Woven plies **homogenised as equivalent orthotropic layers (warp–weft directions)**

HASHIN MODEL PARAMETERS

TABLE 4.1 | Strength parameters used in the Hashin damage model [3].

Property	Symbol	Value [MPa]
Longitudinal tensile strength	X_T	300.34
Longitudinal compressive strength	X_C	339.36
Transverse tensile strength	Y_T	223.65
Transverse compressive strength	Y_C	306.03
Longitudinal shear strength	S_L	54.62
Transverse shear strength	S_T	50.60

TABLE 4.2 | Baseline intralaminar fracture energies used for sensitivity analysis [6].

Damage mode	Symbol	Baseline value [N/mm]
Tensile intralaminar fracture energy	G_f^t	216
Compressive intralaminar fracture energy	G_f^c	136

- Hashin **strength parameters** obtained from literature data for **woven Glass/PPS**
- **No experimental intralaminar fracture energy data available** for the investigated laminate
- **Baseline fracture energies** adopted from **glass/polypropylene** thermoplastic **composite (non-woven / different architectures)**
- Values used as **initial reference for sensitivity analysis and model calibration**

INTRALAMINAR MODEL: VALIDATION

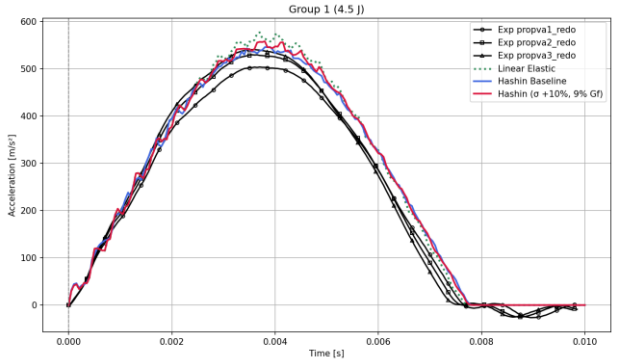


FIGURE 4.8 | *Experimental vs numerical response (4.5 J) - Hashin.*

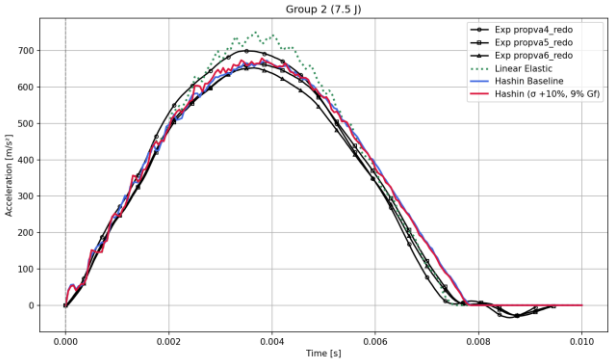


FIGURE 4.9 | *Experimental vs numerical response (7.5 J) - Hashin.*

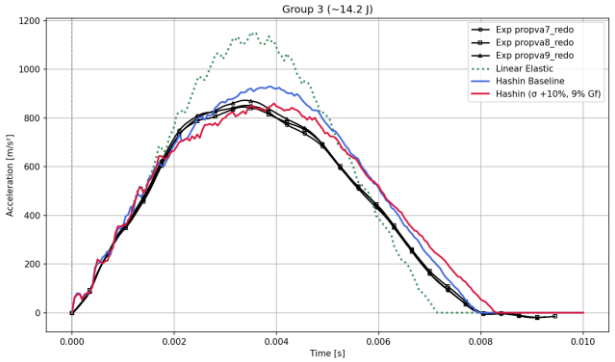


FIGURE 4.10 | *Experimental vs numerical response (14.2 J) - Hashin.*

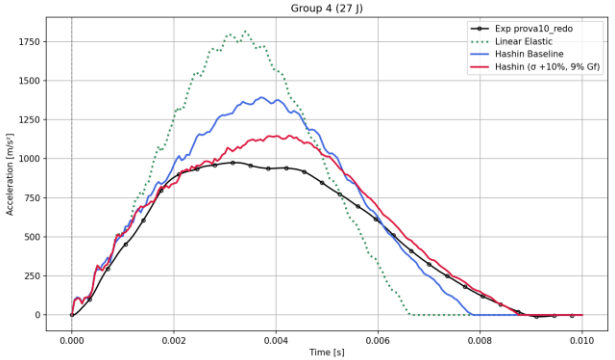
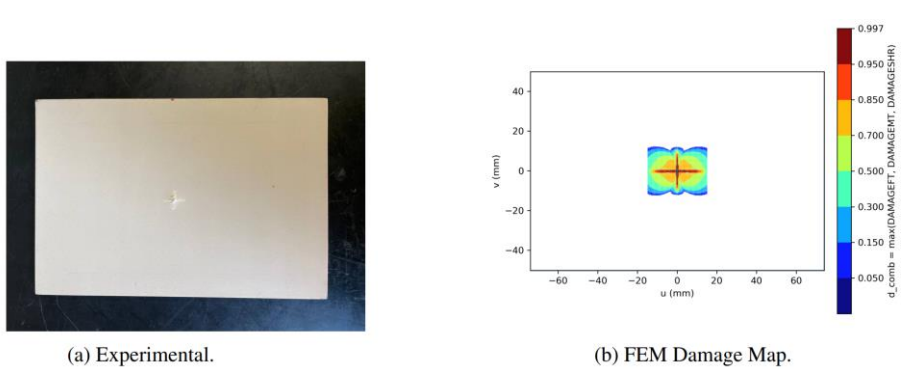


FIGURE 4.11 | *Experimental vs numerical response (27 J) - Hashin.*

INTRALAMINAR MODEL: SENSITIVITY

Table 4.5 | Fracture energy sensitivity (14.2 J).

σ [%]	G_f [%]	L_c^{CAE} [mm]	L_c^{Exp} [mm]	Dev. [%]
–	5.0	22.57	10.33	118.5
–	8.0	13.38	10.33	29.5
–	9.0	10.16	10.33	1.6
–	10.0	7.16	10.33	30.7
+10	**9.0**	**10.62**	**10.33**	**2.8**

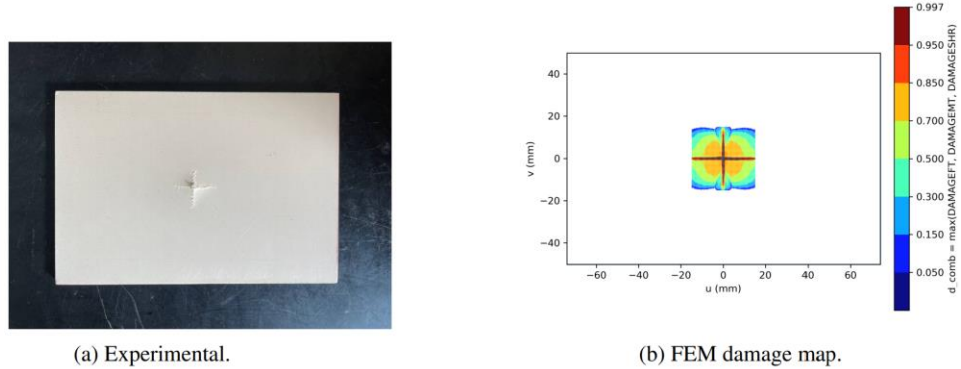


(a) Experimental. (b) FEM Damage Map.

***FIGURE 4.12** | Experimental vs FEM damage pattern (14.2 J).*

Table 4.6 | Fracture energy sensitivity (27 J).

σ [%]	G_f [%]	L_c^{CAE} [mm]	L_c^{Exp} [mm]	Dev. [%]
–	8.0	18.24	19.00	4.0
–	9.0	17.53	19.00	7.7
–	10.0	13.85	19.00	27.1
+10	8.0	20.08	19.00	5.7
+10	**9.0**	**19.38**	**19.00**	**2.0**
+10	9.5	17.99	19.00	5.3



(a) Experimental. (b) FEM damage map.

***FIGURE 4.13** | Experimental vs FEM damage pattern (27 J).*

INTRALAMINAR MODEL: FINAL PROPERTIES

Table 4.7| Baseline vs calibrated intralaminar parameters [3,6].

Property	Baseline value	Reference intralaminar value
Longitudinal tensile strength	$X_T = 300.34$ MPa	$X_T = 330.37$ MPa
Longitudinal compressive strength	$X_C = 339.36$ MPa	$X_C = 373.30$ MPa
Transverse tensile strength	$Y_T = 223.65$ MPa	$Y_T = 246.02$ MPa
Transverse compressive strength	$Y_C = 306.03$ MPa	$Y_C = 336.63$ MPa
Longitudinal shear strength	$S_L = 54.62$ MPa	$S_L = 60.08$ MPa
Transverse shear strength	$S_T = 50.60$ MPa	$S_T = 55.66$ MPa
Tensile intralaminar fracture energy	$G_f^t = 216$ N/mm	$G_f^t = 19.44$ N/mm
Compressive intralaminar fracture energy	$G_f^c = 136$ N/mm	$G_f^c = 12.24$ N/mm

Table 4.8 | Typical intralaminar fracture energy values for woven composites [7-9].

Material system	Architecture	G_f (N/mm)
Carbon/PEEK[a]	Woven laminate	$25 - 40$
Glass/Epoxy[b]	Woven laminate	~ 30
Glass/Elium[c]	Woven laminate	$20 - 35$
Glass/PPS (present work)	5HS woven laminate	19.44

- Parameters selected through **deterministic sensitivity analysis**
- Best agreement obtained for **Gf ≈ 9%** of baseline and **+10% strength**
- Calibrated values reproduce experimental **superficial crack length and morphology**
- Resulting fracture energy values remain within the **literature range**

INTERLAMINAR DELAMINATION: MODEL

Table 4.7| Finite element discretisation including cohesive interfaces.

Element type	Number of elements
Continuum-shell elements (SC8R)	705 280
Cohesive elements (COH3D8)	661 200
Total elements	**1 366 480**

- **Uncertain presence and extent of delamination**
- Delamination modelled using **cohesive elements (COH3D8)**
- Cohesive layers inserted **between laminate plies**
- **Traction–separation law** governs damage initiation and propagation
- Model size increases to **~1.37 million elements**

COHESIVE PARAMETER STUDY

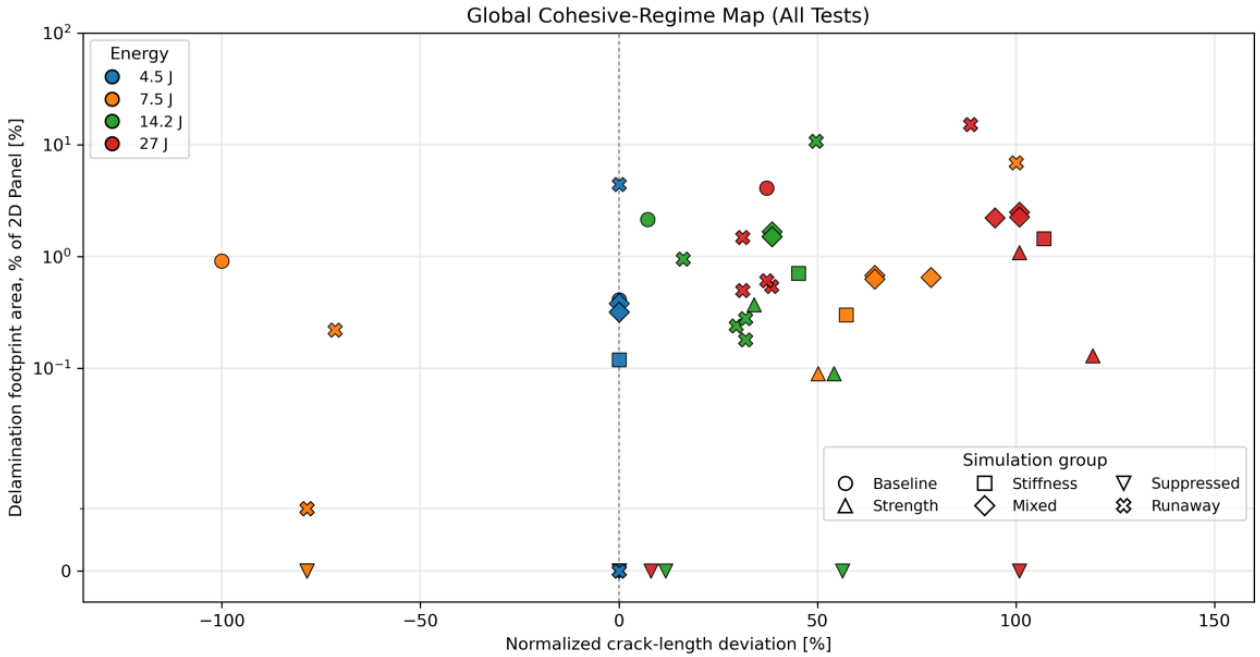


Figure 4.14| Global cohesive parameter regime map obtained from the sensitivity study.

- Global overview of **cohesive model response**
- **Two damage indicators** used:
  - Crack-length deviation
  - Delamination footprint (% panel area)
- Each marker represents one **cohesive parameter configuration**
- **Six response regimes identified:**
  - Baseline
  - Stiffness-modified
  - Stregnth-scaled
  - Mixed
  - Supressed
  - Runaway

COHESIVE PARAMETER INFLUENCE

Table 4.8| Summary of cohesive parameter influence on delamination response.

Cohesive stiffness (K)	
Primary effect	Controls elastic response and energy storage
Sensitivity level	Moderate-High
Effect on footprint	Strong increase; runaway regimes observed
Effect on crack metric	Indirect influence
Interlaminar strength (σ)	
Primary effect	Delays damage initiation
Sensitivity level	Low-Moderate
Effect on footprint	Minor reduction
Effect on crack metric	Weak influence
Fracture energy (G_{Ic}, G_{IIc})	
Primary effect	Governs damage propagation
Sensitivity level	High
Effect on footprint	Strong increase
Effect on crack metric	Limited correlation
BK exponent (η)	
Primary effect	Mode-mix weighting
Sensitivity level	Low
Effect on footprint	Negligible
Effect on crack metric	Negligible

- **Fracture energy** strongly controls **delamination propagation**
- **Cohesive stiffness** governs **interface compliance** and **footprint size**
- **Strength** and **BK exponent** have **secondary influence**

COHESIVE PROPERTY ESTIMATION

Table 4.9| Comparison between baseline and estimated cohesive parameters for the Glass/PPS laminate [7-12].

Parameter	Symbol	Baseline	Estimated
Elastic cohesive stiffness (normal) [MPa/mm]	K_n	4.0×10^3	1.8×10^5
Elastic cohesive stiffness (shear) [MPa/mm]	$K_s = K_t$	1.49×10^3	1.8×10^5
Normal strength [MPa]	σ_{n0}	9.53	60
Shear strength [MPa]	$\tau_{s0} = \tau_{t0}$	5.5	70
Mode I fracture energy [N/mm]	G_{IC}	1.0	1.65
Mode II fracture energy [N/mm]	G_{IIC}	3.0	3.80
BK exponent [-]	η	1.3	1.45

- **Baseline properties** produced **non-physical regimes**
- **Sensitivity** study guided **parameter estimation**
- **Final values consistent with thermoplastic composites**

INTERLAMINAR MODEL: VALIDATION

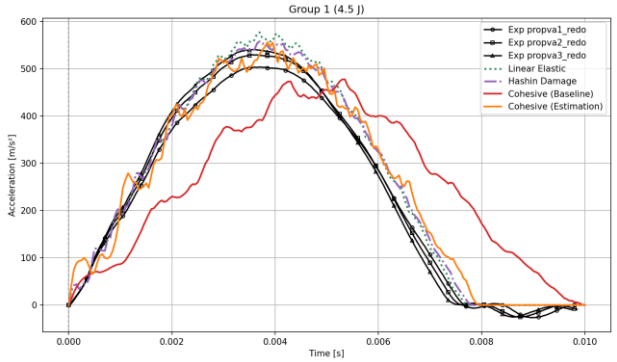


FIGURE 4.8 | *Experimental vs numerical response (4.5 J) - Cohesive.*

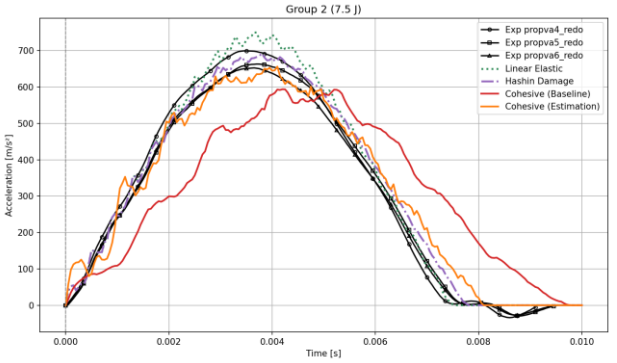


FIGURE 4.9 | *Experimental vs numerical response (7.5 J) - Cohesive.*

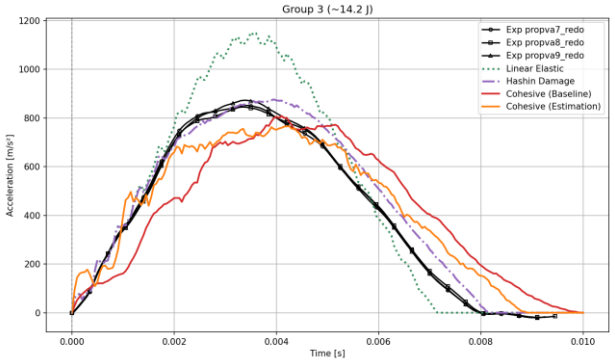


FIGURE 4.10 | *Experimental vs numerical response (14.2 J) - Cohesive.*

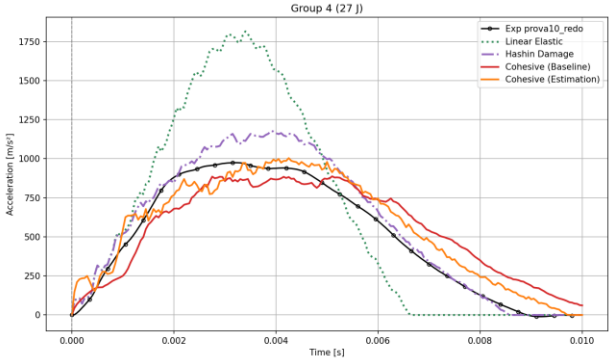


FIGURE 4.11 | *Experimental vs numerical response (27 J) - Cohesive.*

INTERLAMINAR MODEL: METRICS

Table 4.10 | Comparison of predicted delamination metrics for impact energies of 14.2 J and 27 J.

Metric	14.2 J	27 J
Back-face crack		
Crack length (CAE) [mm]	14.31	37.0
Crack length (Exp) [mm]	**10.33**	**19.0**
Delamination area		
Total delamination (SDEG $\geq$ 0.90) [mm^2]	898.2	1554.7
Fully delaminated (SDEG $\geq$ 0.99) [mm^2]	233.7	544.4
Back-face footprint		
Footprint (SDEG $\geq$ 0.90) [mm^2]	118	216
Footprint (SDEG $\geq$ 0.99) [mm^2]	61	105

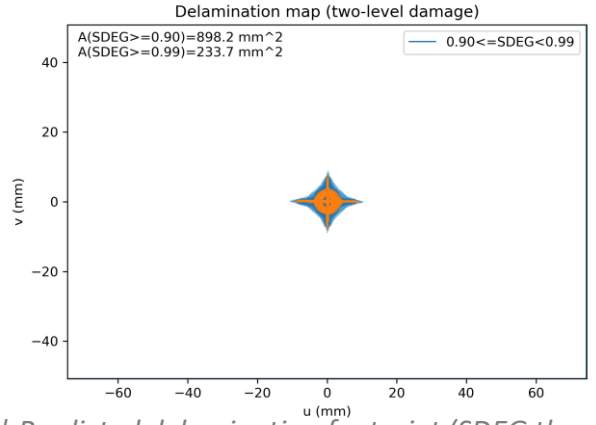


FIGURE 4.12 | Predicted delamination footprint (SDEG thresholds - 14.2 J).

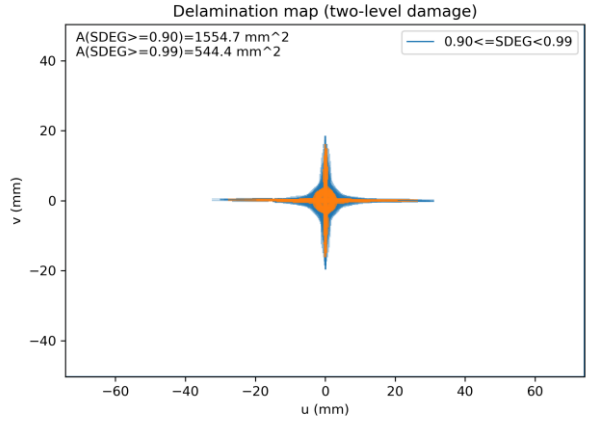


FIGURE 4.13 | Predicted delamination footprint (SDEG thresholds - 27 J).

INTERLAMINAR MODEL: KEY FINDINGS

- **Baseline cohesive** formulation introduced **excessive structural compliance**
- **Estimated cohesive properties preserved laminate stiffness** and matched acceleration response
- **Back-face crack length** systematically **overpredicted** at higher energies

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➢ Motivates the **bi-phasic modelling strategy**

BI-PHASIC MODELLING APPROACH

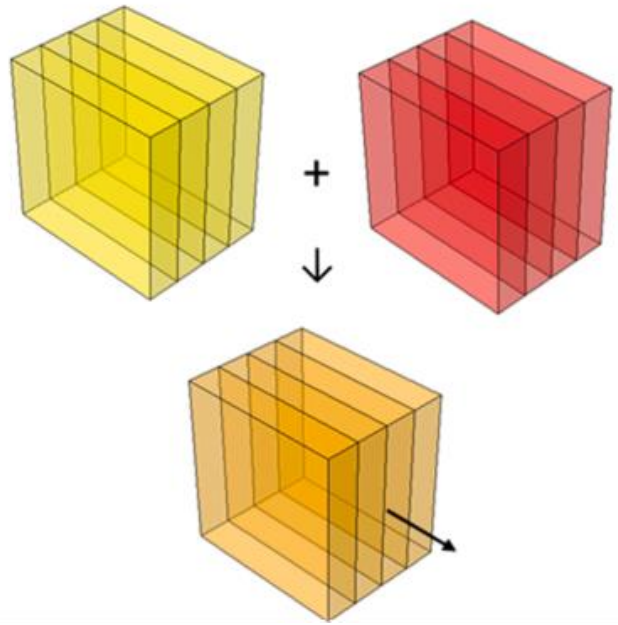


Figure 4.14| Superimposed fibre and matrix phases in the bi-phasic formulation [13].

- **Fibre-dominated phase** represented by continuum-shell elements
- **Matrix-dominated phase** represented by solid elements
- **Superimposed discretisation** with shared nodal connectivity
- Enables representation of **matrix plasticity beneath the impactor**

➢ **Investigates energy dissipation due to matrix plasticity**

INCREMENTAL MODEL DEVELOPMENT

- **Phase I — Elastic bi-phasic baseline**
  Both phases remain elastic to verify preservation of laminate stiffness
- **Phase II — Intralaminar damage activation**
  Hashin damage introduced in the fibre-dominated phase
- **Phase III — Matrix plasticity**
  Plastic deformation introduced in the matrix-dominated solid elements

➢ **Isolates the influence of each physical mechanism**

BI-PHASIC DISCRETISATION STRATEGY

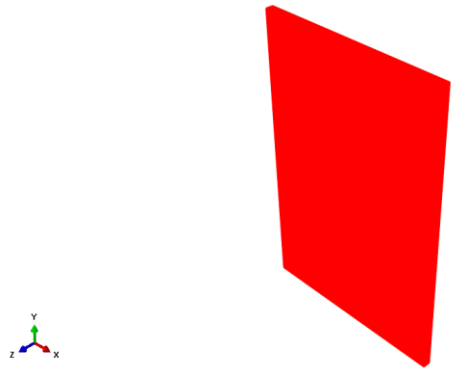


Figure 4.15| Full-panel bi-phasic configuration.

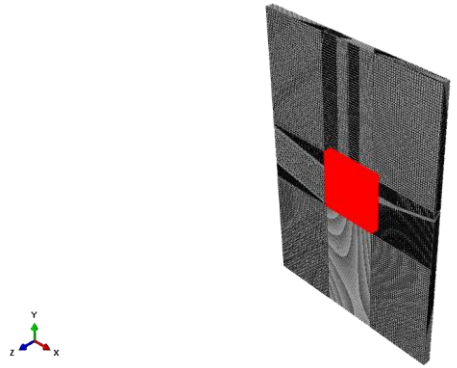


Figure 4.16| Localised impact-zone configuration.

- **Superimposed discretisation** using identical nodal connectivity
- Matrix-dominated **C3D8R solid elements** added to the shell mesh
- **Two geometrical configurations** investigated
  - **Full-panel bi-phasic model**
    Solid elements superimposed across the entire laminate
  - **Localised impact-zone model**
    Solid elements introduced only beneath the impactor
- **Localised configuration reduces solid elements by ~60%** (705k → 270k)

➢ **Evaluates the trade-off between modelling fidelity and computational cost**

ELASTIC MATERIAL PARTITIONING

- Both phases share **identical strain field** (common nodes)
- Laminate stiffness partitioned **(Rule of Mixtures)**

$$V_f = 0{,}5 \qquad V_m = 0{,}5$$

- **Stress and stiffness superposition**

$$\sigma_{lam} = \sigma_{shell} + \sigma_{solid}$$

$$C_{lam} = C_{shell} + C_{solid}$$

➢ **Ensures stiffness consistency and avoids artificial stiffness amplification**

Table 4.11 | Elastic property partitioning for the bi-phasic formulation [3,10].

Property	SC8R Phase [MPa]	C3D8R Phase [MPa]
E_1	21614.5	–
E_2	18254.5	–
E_3	4100.0	–
G_{12}	2864.7	–
G_{23}	2264.7	–
G_{13}	2264.7	–
E (isotropic)	–	2000.0
ν_{12}	0.113	–
ν_{23}	0.32	–
ν_{13}	0.32	–
ν (isotropic)	–	0.36

PHASE I: ELASTIC MODEL VALIDATION

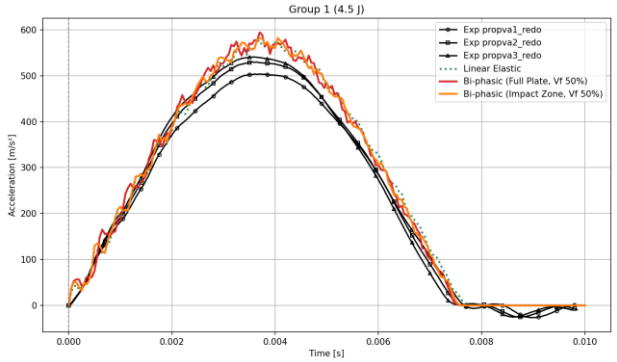


FIGURE 4.17 | *Experimental vs numerical response (4.5 J) – Bi-phasic Elastic.*

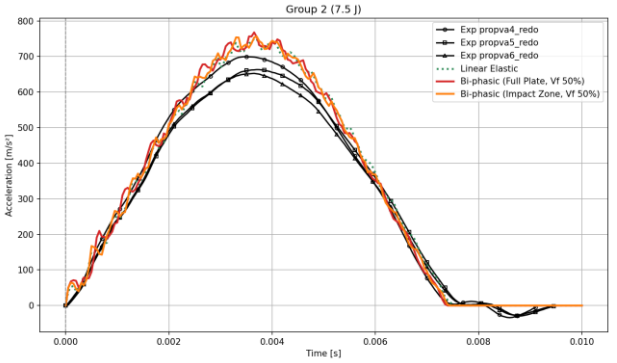


FIGURE 4.18 | *Experimental vs numerical response (7.5 J) - Bi-phasic Elastic.*

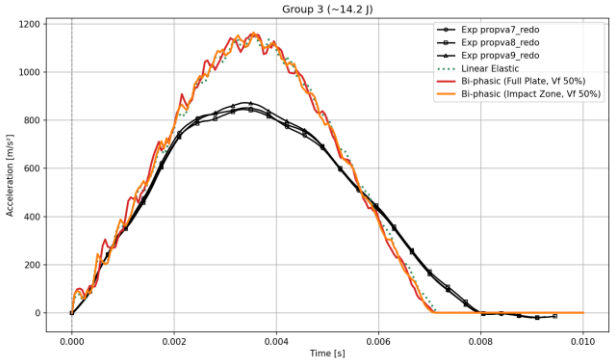


FIGURE 4.19 | *Experimental vs numerical response (14.2 J) - Bi-phasic Elastic.*

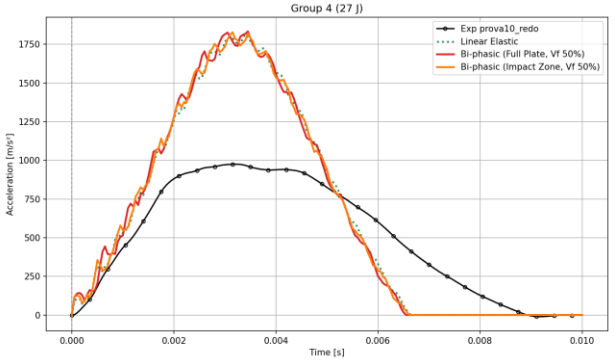


FIGURE 4.20 | *Experimental vs numerical response (27 J) - Bi-phasic Elastic.*

PHASE I: ELASTIC MODEL STIFFNESS CHECK

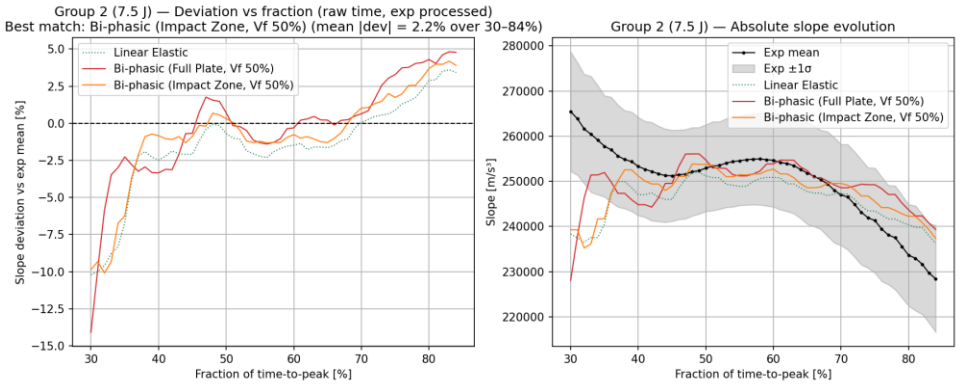


FIGURE 4.21 | 7.5 J — Slope evolution for the bi-phasic elastic model.

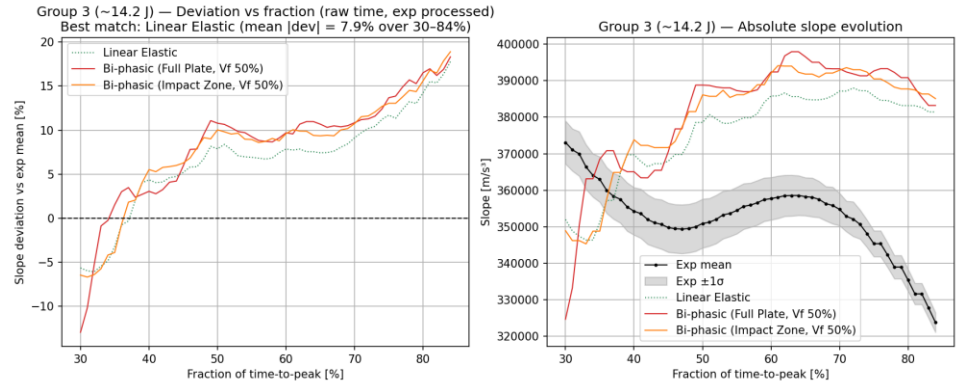


FIGURE 4.22 | 14.2 J — Slope evolution for the bi-phasic elastic model

- **Bi-phasic elastic model reproduces baseline continuum shell response**
- No additional stiffness calibration required

PHASE II: INTRALAMINAR DAMAGE

Hashin intralaminar damage activated in SC8R elements

- Applied to **fibre-dominated phase**
- Implemented in **full-panel and impact-zone configurations**
- **Matrix solid phase kept elastic**

PHASE II: INTRALAMINAR DAMAGE VALIDATION

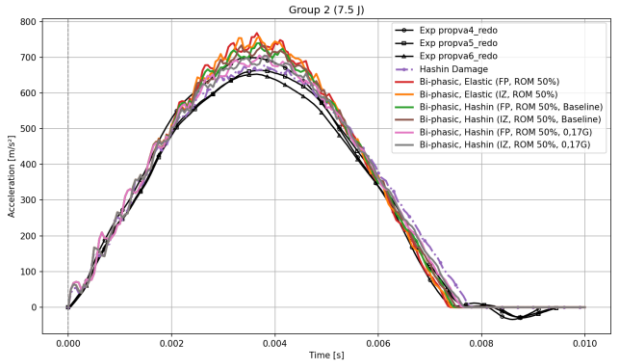


FIGURE 4.23 | *Experimental vs numerical response (4.5 J) – Bi-phasic Hashin.*

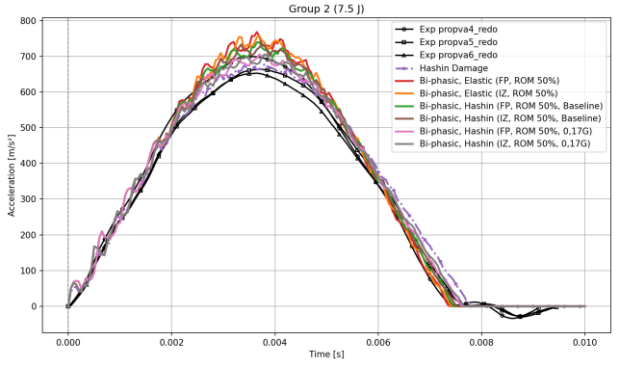


FIGURE 4.24 | *Experimental vs numerical response (7.5 J) - Bi-phasic Hashin.*

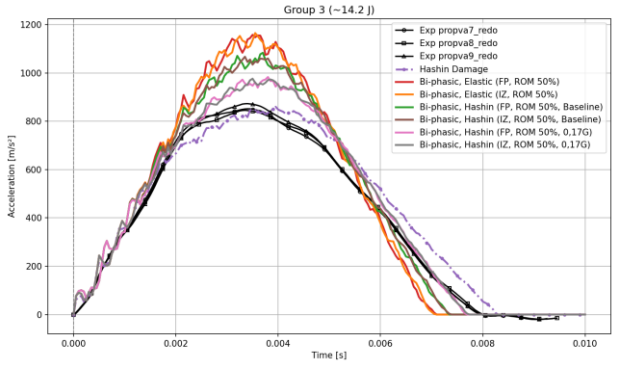


FIGURE 4.25 | *Experimental vs numerical response (14.2 J) - Bi-phasic Hashin.*

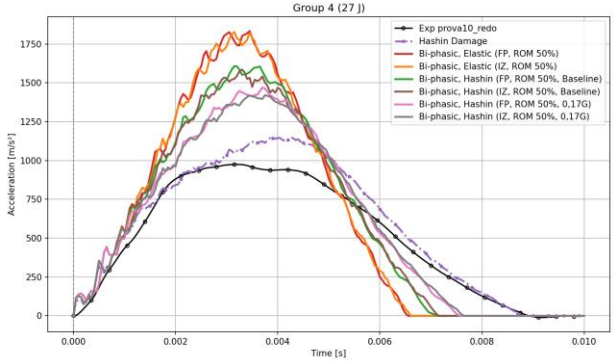
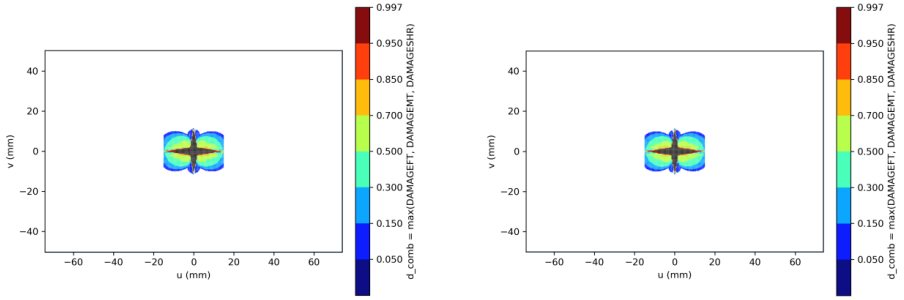


FIGURE 4.26 | *Experimental vs numerical response (27 J) - Bi-phasic Hashin.*

PHASE II: HASHIN SENSITIVITY

Table 4.12 | *Fracture energy sensitivity (14.2 J) – Bi-phasic.*

G_f [%]	L_c^{IZ} [mm]	L_c^{FP} [mm]	L_c^{Exp} [mm]	Dev.
5.0	0.00	0.00	10.33	100%
2.0	5.54	5.76	10.33	46% / 44%
1.7	**9.23**	**9.69**	**10.33**	**11% / 6%**
1.5	20.31	20.77	10.33	97% / 101%

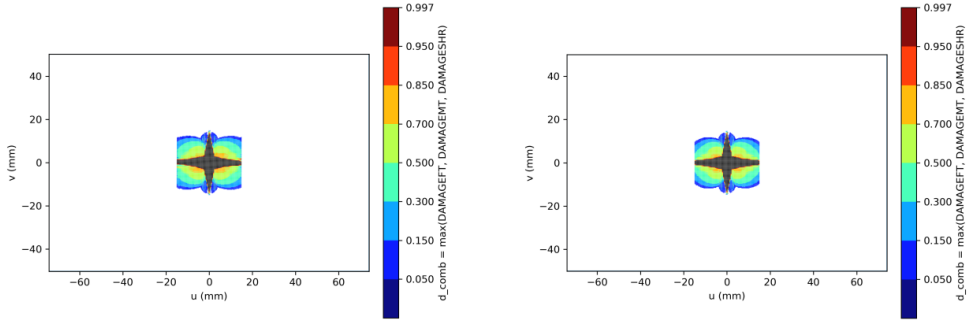


(a) Full Plate (FP). (b) Impact Zone (IZ).

FIGURE 4.27 | *Full Plate (a) vs Impact Zone (b) damage pattern (14.2 J).*

Table 4.13 | *Fracture energy sensitivity (27 J) – Bi-phasic.*

G_f [%]	L_c^{IZ} [mm]	L_c^{FP} [mm]	L_c^{Exp} [mm]	Dev.
5.0	0.00	0.00	19.00	100%
2.0	11.08	12.00	19.00	42% / 37%
1.7	**16.15**	**16.62**	**19.00**	**15% / 13%**
1.5	27.23	27.00	19.00	43% / 42%



(a) Full Plate (FP). (b) Impact Zone (IZ).

FIGURE 4.28 | *Full Plate (a) vs Impact Zone (b) damage pattern (27 J).*

PHASE II: INTRALAMINAR DAMAGE KEY FINDINGS

- **No significant damage differences** between **full-plate** and **impact-zone** models
- Reduced fracture energy values required for calibration are **non-physical numerical artefacts** (19.44 → 3.672 N/mm)
- Acceleration response remains **overly stiff compared to experiments**

➢ **Motivates the introduction of matrix plasticity**

PHASE III: MATRIX PLASTICITY

Perfect plastic behaviour assumed for the matrix phase

- Plasticity implemented in **matrix-dominated solid elements**
- Only **14.2 J and 27 J** analysed to reduce computational cost
- **Impact-zone configuration adopted** for parametric studies

Table 4.14 | Mechanical properties adopted for the PPS matrix phase [10].

Property	Symbol	Value
Elastic modulus	E	2000 MPa
Poisson ratio	ν	0.36
Tensile strength	σ_t	102 MPa
Compressive strength	σ_c	148 MPa

PHASE III: PLASTICITY MODELS INVESTIGATED

Von Mises

- Pressure-independent plasticity
- Yield governed by **deviatoric stress**
- **Isochoric plastic flow**

Table 4.15 | Von Mises Plasticity Parameters [10].

Parameter	Value
Yield stress σ_y	148 MPa
Hardening	Perfect plasticity

Drucker-Prager

- Pressure-dependent plasticity
- Yield depends on **hydrostatic + deviatoric stress**
- Suitable for **polymer matrix behaviour**

Table 4.16 | Drucker-Prager Plasticity Parameters [10].

Parameter	Value
Friction angle β	28.9°
Cohesion d	120.8 MPa
Dilation angle ψ	0°
Deviatoric parameter K	1.0

Crushable Foam

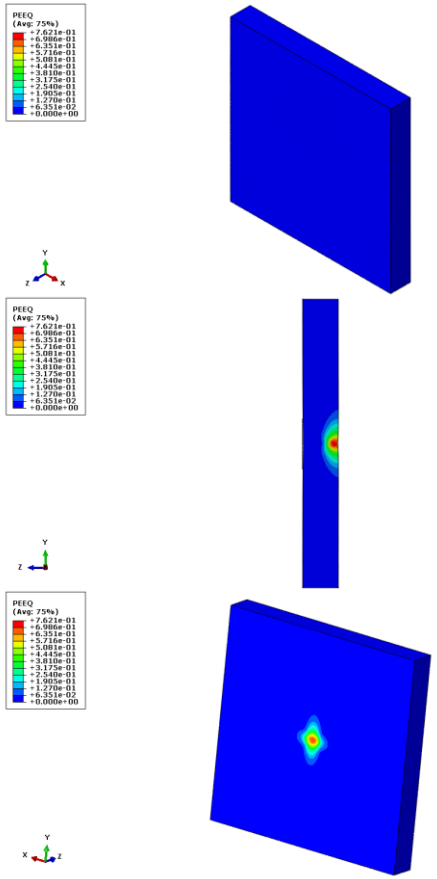
- Plasticity with **volumetric compaction**
- Designed for **compressible materials**
- Captures **densification under compression**

Table 4.17 | Crushable Foam Plasticity Parameters [10].

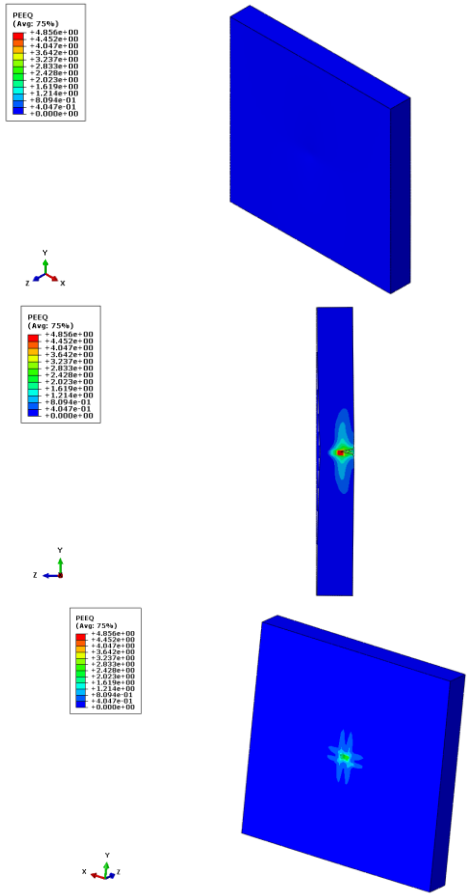
Parameter	Value
Compressive yield stress σ_c	148 MPa
Compression yield ratio k	1.0
Hydrostatic yield ratio k_t	0.689

PHASE III: PEEQ COMPARISON

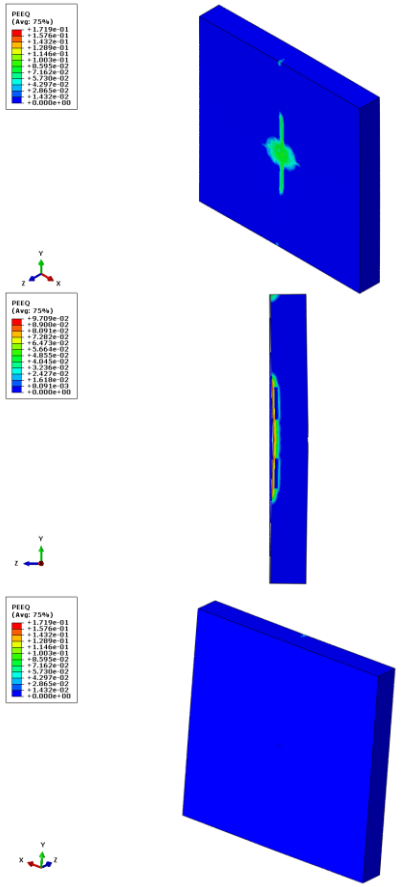
Von Mises



Drucker-Prager



Crushable Foam



PHASE III: PLASTICITY MODEL ASSESSEMENT

Von Mises

- Plastic deformation develops mainly on the **tensile back face**
- **No plasticity beneath the impactor**
- Pressure-independent yielding ignores **hydrostatic compression**
- **Unable to reproduce indentation behaviour**

Drucker-Prager

- Plastic deformation still concentrated on the **tensile back face**
- **No plasticity beneath the impactor**
- Parameter variations produce **minor changes only**
- **Indentation behaviour not captured**

Crushable Foam

- Plastic deformation develops **beneath the impactor**
- Formation **of localized plastic zone in the impact region**
- Through-thickness **compressive plastic band**
- **Indentation behaviour successfully reproduced**

PHASE III: PLASTICITY VALIDATION

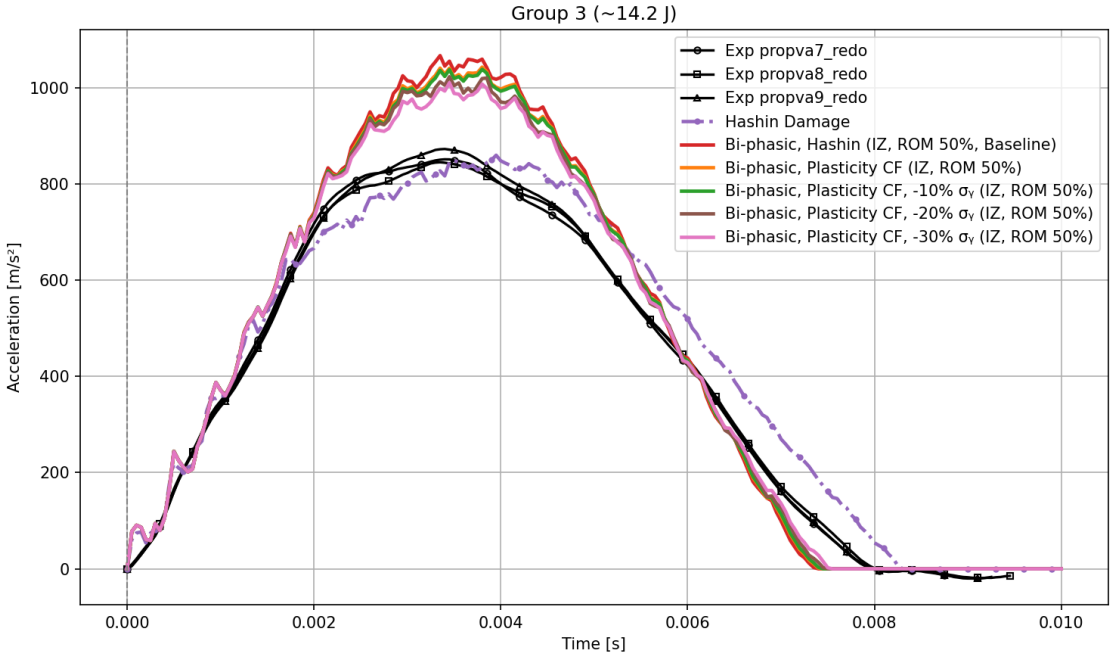


FIGURE 4.29 | *Experimental vs numerical response (14.2 J) - Bi-phasic Plastic.*

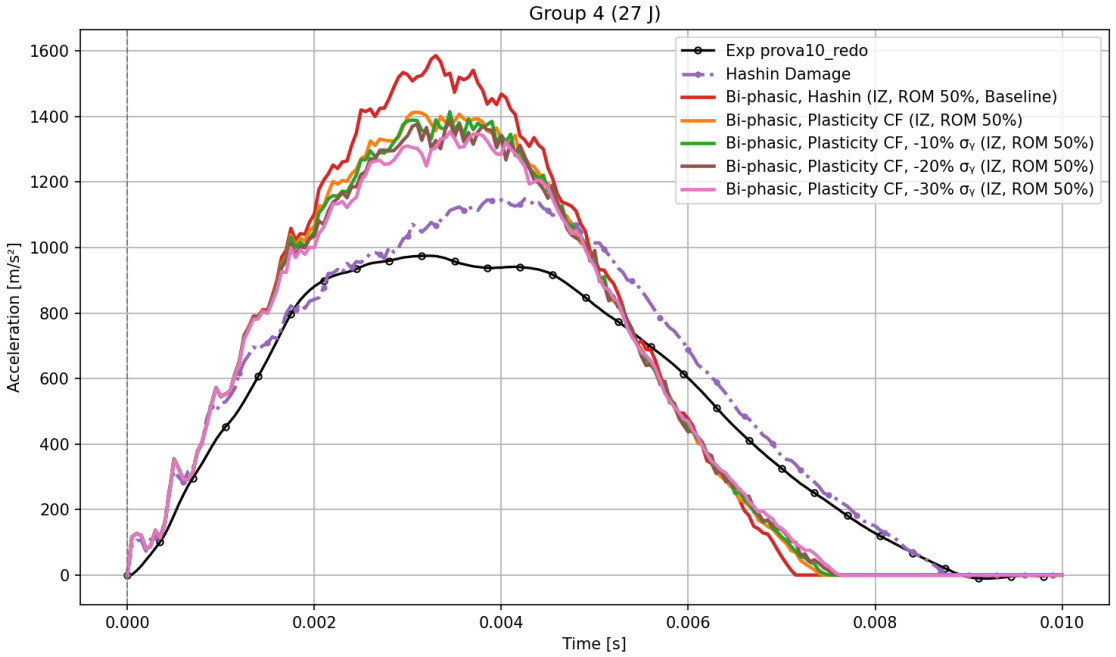


FIGURE 4.30 | *Experimental vs numerical response (27 J) - Bi-phasic Plastic.*

PHASE III: YIELD STRESS SENSITIVITY

Table 4.18 | Yield Stress sensitivity (14.2 J) .

σ_y variation	L_c^{CAE} [mm]	L_c^{Exp} [mm]	$Indentation_{CAE}$ [mm]	$Indentation_{Exp}$ [mm]
Baseline	0.00	10.3	3.92	7.40
−10%	0.00	10.3	4.38	7.40
−20%	0.00	10.3	4.84	7.40
−30%	0.00	10.3	5.54	7.40

Table 4.19 | Yield Stress sensitivity (27 J) .

σ_y variation	L_c^{CAE} [mm]	L_c^{Exp} [mm]	$Indentation_{CAE}$ [mm]	$Indentation_{Exp}$ [mm]
Baseline	9.23	19	6.00	9.00
−10%	10.62	19	5.77	9.00
−20%	12.00	19	5.30	9.00
−30%	12.92	19	6.00	9.00

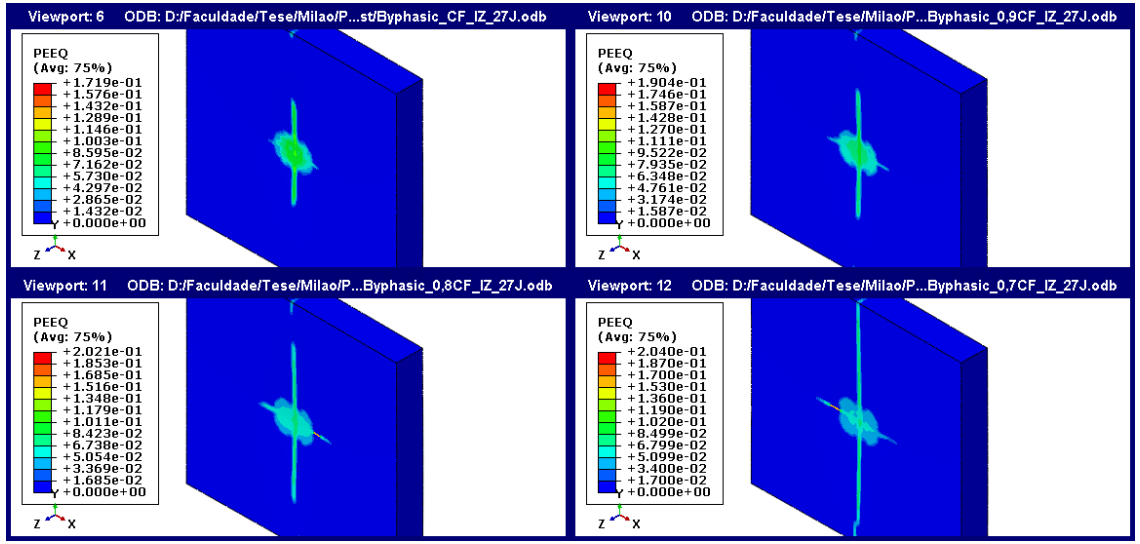


FIGURE 4.31 | PEEQ distribution for different matrix yield strengths (27 J).

- **Plasticity controls indentation at 14.2J;** cracking governed by laminate damage
- At **27J,** lower σ_y **increases crack length**, while **indentation changes little**
- Reduced σ_y **expands the plastic zone**, forming **cross-shaped arms**
- Cross-shaped arms resemble **wrinkling/creases, not observed experimentally → model limitation**

PHASE III: HASHIN FRACTURE ENERGY EFFECT

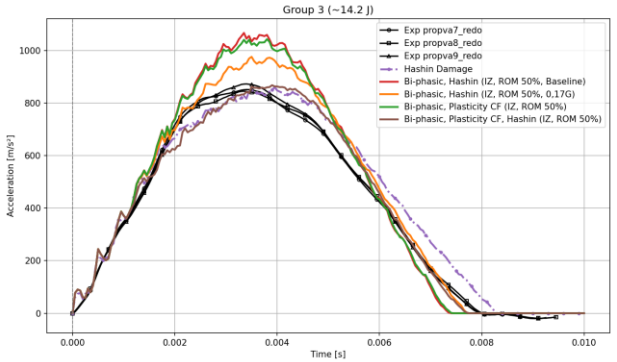


FIGURE 4.32 | Intralaminar Influence on Plasticity (14.2J).

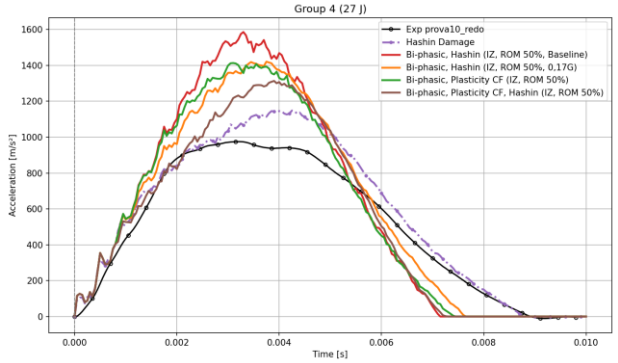


FIGURE 4.33 | Intralaminar Influence on Plasticity (27J).

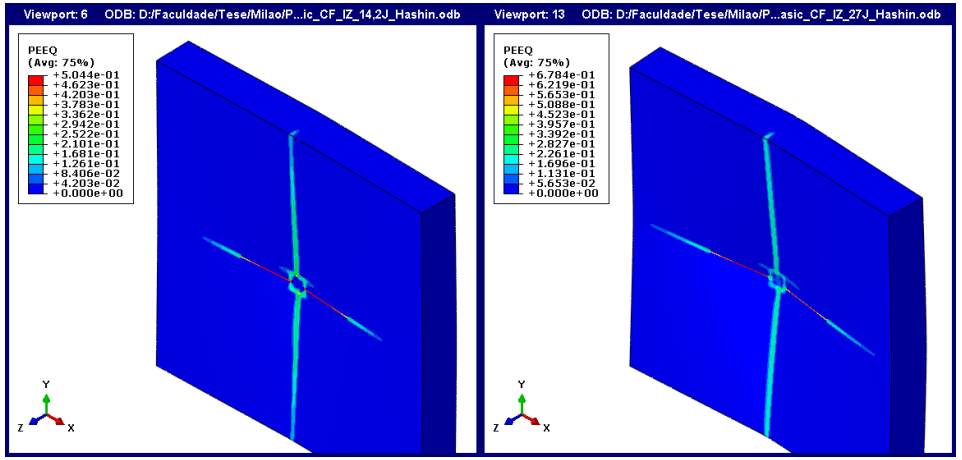


FIGURE 4.34 | PEEQ distribution for the calibrated intralaminar fracture energy.

- **Fracture energy calibrated in the previous bi-phasic sensitivity study** was reintroduced to assess its interaction with matrix plasticity
- These values **trigger premature intralaminar damage**, producing **excessive crack growth**
- **Indentation is strongly suppressed** (~2 mm), indicating **damage dominates energy dissipation**
- **Large cross-shaped plastic bands** appear, resembling **wrinkling not observed experimentally**

ULTRASONIC INSPECTION

- Post-impact **ultrasonic C-scan inspection**
- Identification of **internal delamination**
- Quantification of **delamination footprint**
- Assessment of **damage evolution with impact energy**

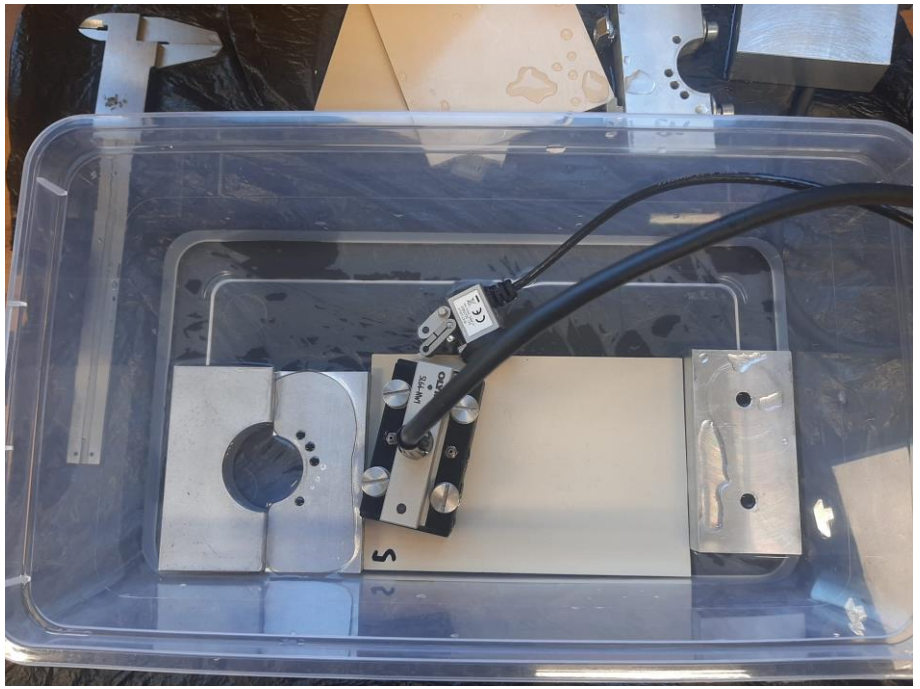


FIGURE 5.1 | *Ultrasonic immersion inspection setup.*

DELAMINATION AREA EVOLUTION

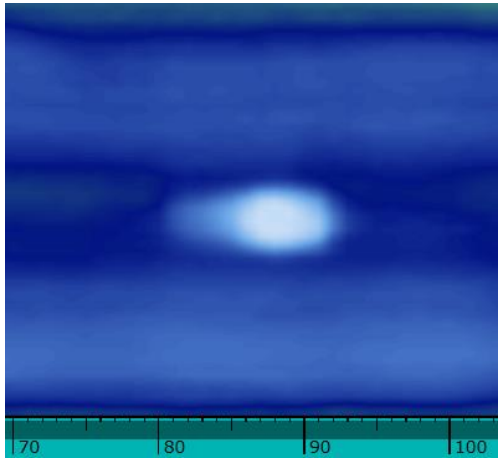


FIGURE 5.2 | C-scan map – 7.5 J impact.

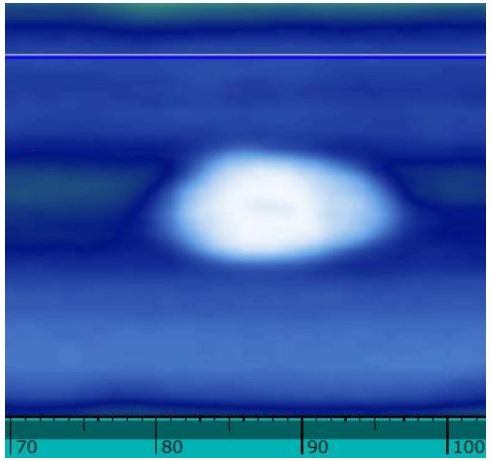


FIGURE 5.3 | C-scan map – 14.2 J impact.

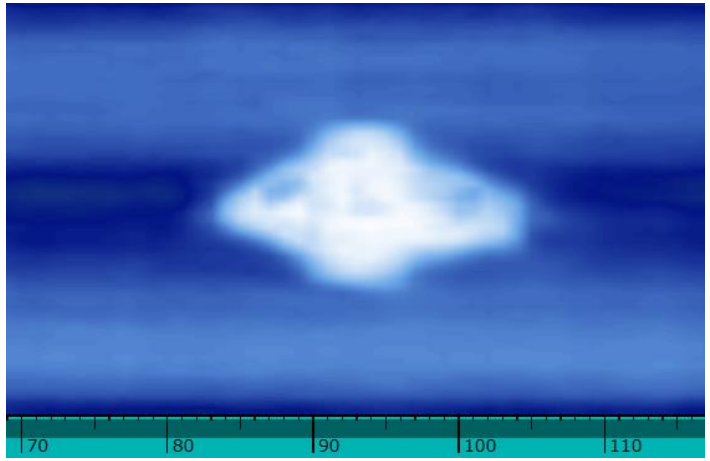


FIGURE 5.4 | C-scan map – 27 J impact.

Table 5.1 | Delamination area vs impact energy.

Impact Energy [J]	Mean Area [mm^2]	Std. Dev. [mm^2]	Rel. Dev. [%]
7.5	19.93	2.88	14.4
14.2	80.04	4.93	6.2
27	142.70	13.31	9.3

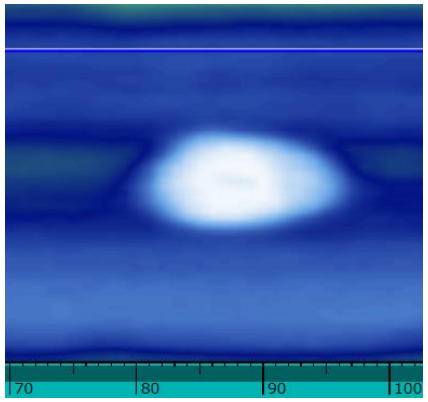
ENERGY DISSIPATION MECHANISMS

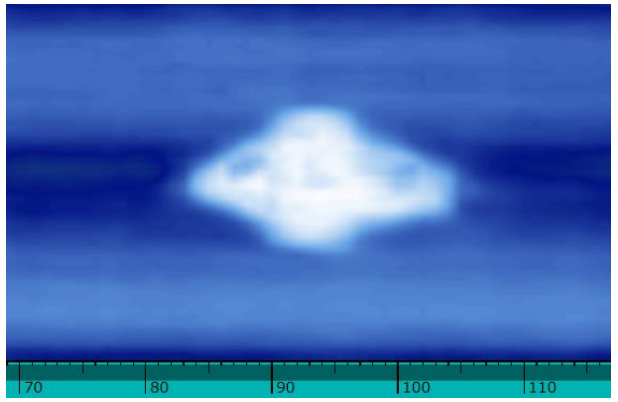
- Ultrasonic inspection confirms **significant interlaminar delamination**
- Impact response governed by **multiple coupled mechanisms**
- Energy dissipated through:
  - **Intralaminar fibre fracture**
  - **Interlaminar delamination**
  - **Matrix plastic deformation**

Table 5.2 | Main energy dissipation mechanisms identified during impact.

Mechanism	Evidence	Model
Intralaminar fracture	Superficial back-face crack	Hashin damage
Delamination	C-scan inspection	Cohesive elements
Matrix plasticity	Indentation	Bi-phasic plasticity

FRACTURE ENERGY SENSITIVITY





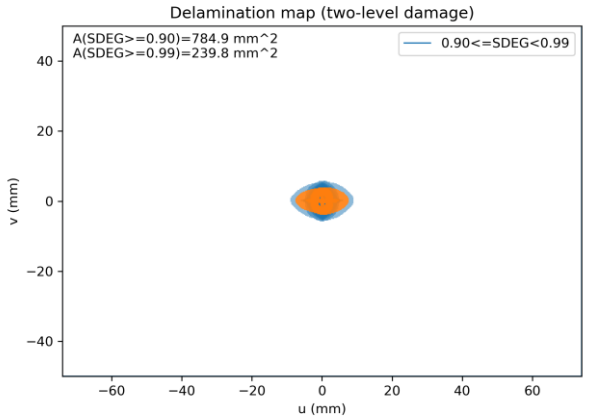


FIGURE 5.5 | 14.2 J — C-scan and CAE delamination map.

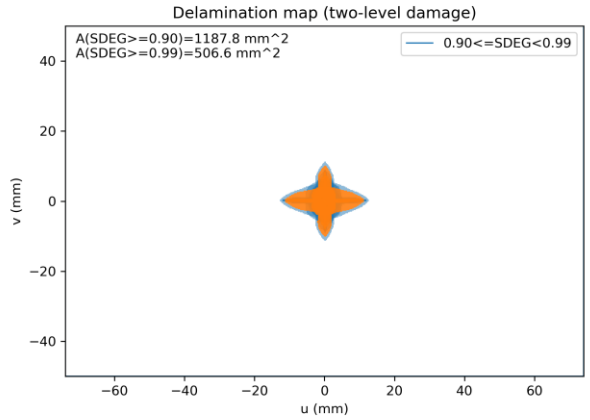


FIGURE 5.6 | 27 J — C-scan and CAE delamination map.

Table 5.3 | Comparison of delamination area and back-face crack (Exp vs CAE).

	14.2 J	27 J
Delamination (NDT) [mm^2]	**80**	**143**
Delamination (CAE) [mm^2]	**90**	**165**
Back-face crack (Exp) [mm]	10.3	19
Back-face crack (CAE) [mm]	6	16

- Increased intralaminar fracture energy (**20 → 30 N/mm**)
- Improved prediction of **delamination area and damage morphology**
- Indicates redistribution of energy between **fracture and delamination**

GLOBAL RESPONSE VALIDATION

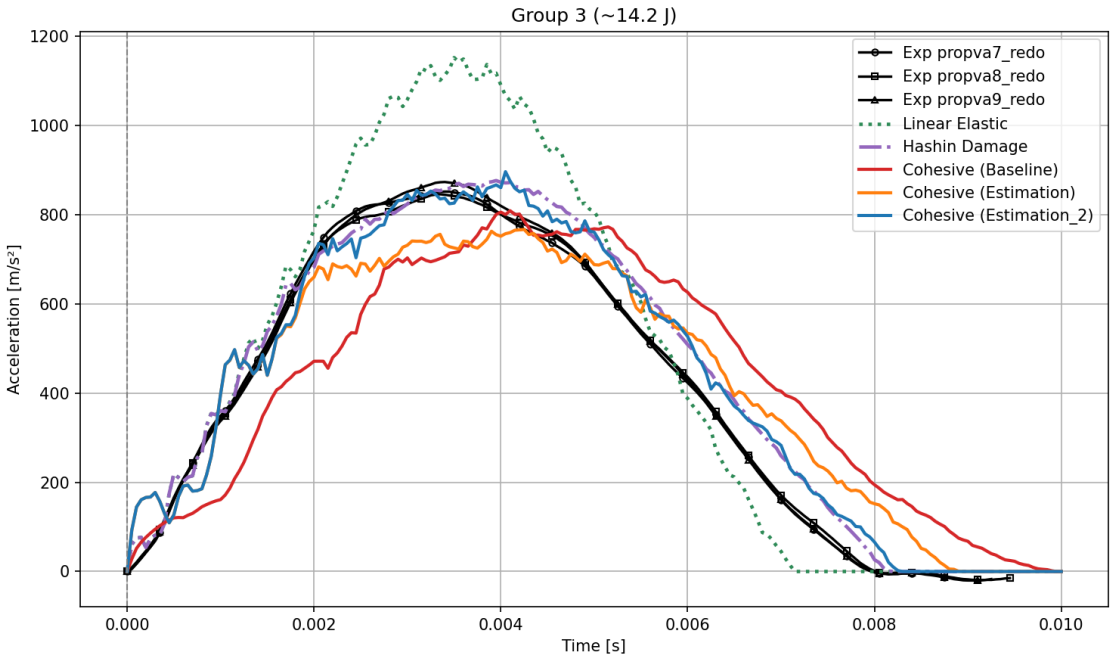


FIGURE 5.7 *| Experimental vs numerical response (14.2 J) – Updated model.*

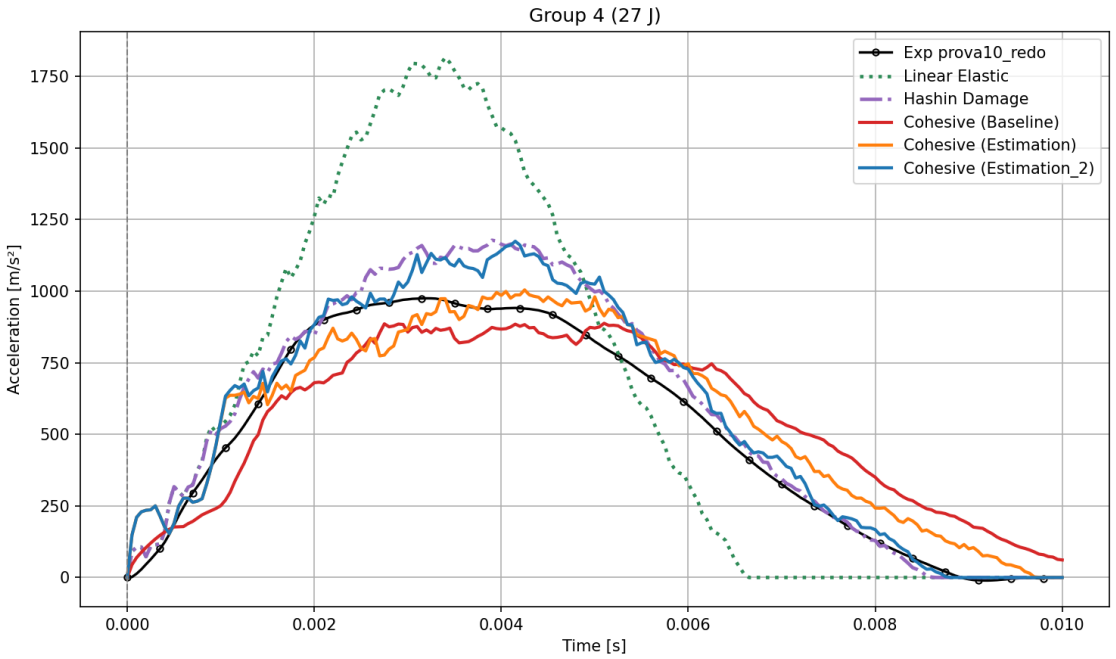


FIGURE 5.8 *| Experimental vs numerical response (27 J) - Updated model.*

SUMMARY OF THE WORK

- Investigation of **low-velocity impact behaviour** in Glass/PPS thermoplastic laminates
- Development of a **finite element modelling framework** using Abaqus/Explicit
- Progressive modelling strategy including:
  - **Intralaminar Damage (Hashin 2D)**
  - **Interlaminar delamination (Cohesive elements)**
  - **Matrix plastic deformation (Bi-phasic approach)**
- **Ultrasonic C-scan inspection** to characterize post-impact delamination
- Development of **Python tools** for automated extraction of damage metrics

MAIN FINDINGS

- Impact damage severity **increases with impact energy**
- **Ultrasonic C-scan inspections** reveal **significant interlaminar delamination**, increasing with impact energy
- **Intralaminar fibre fracture** governs crack formation in the laminate
- Impact response governed by **coupled damage mechanisms**:
  - **Intralaminar fracture**
  - **Interlaminar delamination**
  - **Matrix deformation beneath the impactor**
- Hashin intralaminar model provides **accurate crack prediction with low computational cost**
- Delamination prediction is **sensitive to fracture energy definition**

KEY MODELLING INSIGHTS

- Calibrated intralaminar fracture energy values were **slightly lower than literature values for woven composites**
- In simplified models**, intralaminar fracture energy may act as an effective energy absorption parameter**, masking other damage mechanisms
- **Lower intralaminar fracture energy values may lead to partial suppression of interlaminar delamination**, as confirmed by sensitivity analysis
- Impact response governed by the interaction between:
  - **Intralaminar fibre fracture**
  - **Interlaminar delamination**
- Accurate impact modelling requires **explicit representation** and **correct balance of multiple dissipation mechanisms**
- **Updated cohesive properties**, including higher intralaminar fracture energy, **provide a more realistic material response, consistent with literature values**

LIMITATIONS AND FUTURE WORK

LIMITATIONS

- Some material parameters estimated from **literature values**
- Simplified representation of complex **damage mechanisms**
- Absence of full **3D experimental characterization (µCT)**

FUTURE WORK

- Experimental characterization of **interlaminar fracture energies (DCB / ENF)**
- Improved characterization of **PPS matrix plastic behaviour**
- Use of **µCT imaging** for detailed internal damage validation
- Use **of probabilistic methods (e.g., Monte Carlo simulations)** to assess the **influence of modelling parameters** and their uncertainty on impact response

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THANK YOU FOR YOUR ATTENTION!

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